

Final Programming Report Specification

Statistical / Machine Learning Course

1 Project Overview

This is a two-week team programming report. Each team should consist of **three students**. The project is designed to evaluate your ability to apply statistical learning methods covered in the course to a real dataset. The project includes both an oral presentation and a written report.

Each team will choose one dataset from the suggested options listed in Section 7. Different teams may choose the same dataset. However, each team must define its own analysis objective clearly and should not simply reproduce a standard online analysis.

Each team must (1) clearly state the question it wants to answer, describe the dataset, (2) explain the data processing steps, justify the selected methods, and (3) present the final results with appropriate visualizations.

2 Important Dates and Reporting Arrangement

- By **May 23, 23:59**, each team must email me (walin@math.ncu.edu.tw) with the following information:
 - team member names and student ID;
 - selected dataset;
 - tentative project topic or analysis goal.
- On **May 27**, I will announce online the presentation order for **June 3** and **June 10**.
- Each team will have approximately **30 minutes** for the oral presentation.
- Each presentation will be followed by approximately **10 minutes** of discussion, during which other students and me may ask questions. Team members are expected to answer these questions.

3 Oral Presentation Requirement

Each team must prepare slides for the oral presentation. The slides should be clear, well organized, and focused on the analysis rather than on raw code.

The oral presentation must include the following three major components:

1. **Data Processing and Descriptive Statistical Presentation**
 - Describe the dataset and the variables used in the analysis.

- Explain how missing values, outliers, categorical variables, scaling, transformations, or feature construction were handled.
- Present meaningful descriptive statistics and exploratory visualizations.

2. Method Details

- Explain which statistical learning methods were used.
- Explain why each method was selected.
- Explain how the method matches the dataset and the analysis goal.
- Discuss tuning parameters, model selection, validation, and any important implementation details.

3. Results and Visualization

- Present model performance or exploratory findings clearly.
- Compare the selected methods.
- Include appropriate tables and visualizations.
- Interpret the results in relation to the original analysis goal.

Each of these three major components will be evaluated separately. **Individual scores will be assigned according to the work percentages reported by the team.** Therefore, each team member's work percentage must be filled in accurately and responsibly.

If a team member is not contributing responsibly and communication within the team has become difficult, students should email me. I will investigate the situation and handle the matter when necessary.

4 Written Report Requirement

Each team must also submit a written report to me.

The written report must follow a report format similar to the midterm programming assignment. The final submission must be a **single PDF file**. Separate scripts, notebooks, data files, or supplementary code files will not be accepted.

The report should not be a code listing. Code should not be the main content of the report. Instead, the report should focus on the analysis, methodology, results, and interpretation. If code fragments are necessary for explanation, they should be short and directly relevant.

The written report must contain the following sections.

4.1 Research Goal

State the main question your team wants to answer.

The goal must be specific. For example, do not write only:

We analyze the Wine Quality data.

Instead, write something like:

We aim to predict whether a wine receives a high quality score based on its physicochemical measurements, and to identify which variables are most important for this prediction.

Your goal may be predictive, exploratory, or both. However, it must be concrete enough that your results can be evaluated.

4.2 Dataset Description

Describe the dataset in your own words. Include:

- the source of the data;
- the number of observations;
- the number and types of variables;
- the response variable, if your project is supervised;
- the main features used in the analysis;
- any missing values, outliers, imbalance, or data quality issues.

4.3 Data Processing and Exploratory Analysis

Explain every important data-processing step. This may include:

- removing or imputing missing values;
- transforming variables;
- standardizing numerical variables;
- creating dummy variables for categorical predictors;
- constructing new features;
- splitting the data into training, validation, and test sets;
- handling class imbalance;
- choosing which variables to include or exclude.

The report must include meaningful visualizations. These visualizations should support the analysis rather than merely display the data.

4.4 Method Details

Apply at least three statistical learning methods. At least one method should be a simple baseline model. At least one method should be a more flexible model.

Possible methods include:

- linear regression or logistic regression;
- ridge, lasso, or elastic net;
- polynomial regression, splines, or generalized additive models;
- classification trees or regression trees;

- random forests;
- boosting;
- support vector classifiers or kernel SVMs;
- principal components analysis;
- K-means clustering;
- hierarchical clustering;
- neural networks, CNNs, or RNNs, if appropriate.

For each method, explain:

- why the method is appropriate for your objective;
- how the method matches the dataset and topic;
- what inputs were used;
- what tuning parameters were considered;
- how the tuning parameters were selected;
- what validation strategy was used.

4.5 Results and Visualization

Present the results clearly.

For supervised learning projects, report appropriate performance measures, such as:

- test MSE or RMSE for regression;
- test accuracy, sensitivity, specificity, precision, recall, F1 score, AUC, or a confusion matrix for classification.

For unsupervised learning projects, report appropriate exploratory evidence, such as:

- PCA loading interpretation;
- proportion of variance explained or scree plot;
- cluster visualization;
- dendrogram interpretation;
- within-cluster variation;
- stability or sensitivity of clusters under different choices.

You must compare the methods, not just report them separately.

4.6 Interpretation and Discussion

Explain what your results mean in relation to the original research goal. Discuss:

- which method performed best and why;
- whether the results are stable or sensitive to modeling choices;
- which variables appear important;
- whether the findings are practically or scientifically meaningful;
- limitations of the analysis;
- what you would do next if more time or data were available.

5 Team Contribution Statement

At the end of both the slides and the written report, include a team contribution statement. The statement must report how much each member contributed to each of the three major components.

The three components are:

1. Data Processing and Descriptive Statistical Presentation;
2. Method Details;
3. Results and Visualization.

For each component, state the percentage contribution of each team member. Each row must sum to 100%.

Use the following format:

Component	Student A	Student B	Student C
Data Processing and Descriptive Statistical Presentation	40%	40%	20%
Method Details	30%	50%	20%
Results and Visualization	30%	30%	40%

Do not fill in these percentages casually. Since each component is evaluated separately, individual scores will be assigned based on the reported contribution percentages. False or careless contribution reporting may affect individual grading.

If the division of work requires clarification, add a short written explanation below the table.

6 Minimum Technical Requirements

Each team must include:

- one clearly stated analysis goal;
- one reproducible data-loading procedure;

- meaningful descriptive statistics and exploratory visualizations;
- at least three statistical learning methods;
- one clear validation or evaluation strategy;
- one final method comparison;
- one interpretation section;
- one team contribution table.

The report should read as a coherent data analysis report. It should not be a sequence of screenshots, raw outputs, or disconnected figures.

7 Suggested Dataset Options

Each team may choose one of the following datasets. Other datasets may be used only with instructor approval.

Dataset	Main Task Type	Why It Is Suitable
Wine Quality	Regression or classification	Predict wine quality from physicochemical measurements. Suitable for regression, binary classification after defining “high quality”, variable importance, trees, random forests, boosting, SVMs, and regularized regression.
Breast Cancer Wisconsin Diagnostic	Binary classification	Predict whether a tumor is malignant or benign using real-valued features computed from digitized images of fine needle aspirates. Suitable for logistic regression, lasso, SVM, random forest, boosting, PCA, and model interpretation.
Adult Census Income	Binary classification	Predict whether income exceeds 50K per year using demographic and employment-related variables. Suitable for logistic regression, trees, random forests, boosting, SVMs, and discussion of categorical variables and class imbalance.
Bank Marketing	Binary classification	Predict whether a client subscribes to a term deposit after a direct marketing campaign. Suitable for classification, imbalanced data, variable importance, and comparison of interpretable and flexible models.
Ames Housing	Regression	Predict house sale price using housing characteristics. Suitable for feature engineering, missing value handling, lasso or ridge regression, tree-based models, boosting, and nonlinear effects.

Dataset	Main Task Type	Why It Is Suitable
Online Retail	Clustering, segmentation, or prediction	Transactional data from an online retail store. Suitable for customer segmentation, RFM-style features, clustering, PCA, and possibly predicting customer spending or repeat purchase behavior.
Palmer Penguins	Classification or clustering	Classify penguin species or explore clustering using body measurements. Suitable for a smaller but clean project involving visualization, classification, PCA, and clustering.

8 Submission Format

Submit one PDF file only for the written report.

The PDF must include:

- written explanations;
- tables and figures;
- model outputs needed to support the conclusions;
- a team contribution table.

No separate R scripts, Python scripts, notebooks, or data files will be accepted. Students may use R, Python, or another suitable programming language, but the final PDF must be self-contained as a report.